NAVER Cloud

Starting a Business with NAVER Cloud

Version 1.1

Update 2021.3

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Introduction to NAVER Cloud

NAVER Cloud strives to solve IT needs and to make businesses succeed.

NAVER has become a part of our lives, establishing itself to be the most convenient and user-friendly technology putting its customers first. We share NAVER's technology, experience, and knowledge that has been compiled over the years with a myriad of companies and creators. And through this, we work to enrich the IT ecosystem and become the Korea's leading cloud company that expands globally.

We will continue do our best to support the IT services and to operate them securely so that the numerous customers and partners of NAVER, LINE, SNOW, and other NAVER affiliates can easily utilize the various services offered.

Company Name	NAVER Cloud Corp.
Foundation Date	May 1, 2009
CEO	Weongi Park
Main Business	Cloud and IT services
Total Revenue	492,580,510,000 won (Dec. 2019 IFRS Independent)
Headquarters	[Headquarters] Krafton Tower, Bundang Naerok-ro 117 Bundang-gu, Seongnam-si, Gyeonggi-do [Gangnam Office & Training Center] Gangnam N Tower, Taehaeran-ro 129 Gangnam-gu, Seongnam-si, South Korea
Phone Number	1566-3880 / 031-784-1800
Homepage	www.navercloudcorp.com (NAVER Cloud) www.ncloud.com (NAVER Cloud Platform)

Corporate Identity

NAVER Cloud

We will become the top global technology service that shares NAVER's technology and knowledge to customers and grow as leader in expanding the IT ecosystem through solving companies and creators' concerns.

Slogan

Break Through with NAVER Cloud

Mission

Customer Centric

We provide cloud-based solutions customized to each industry and service at a rapid pace.

Integrated Platform

We consolidate NAVER's technology and diverse business solutions all onto one platform.

Global Leadership

We lead the cloud market with our international regions and with our successful global service operation experience.

IT Ecosystem

We will continue with our various efforts to support and grow with architects of the IT ecosystem.

History

NAVER Cloud contains the experience and knowledge from the NAVER, the rapidly growing Korea's #1 IT platform, NAVER, from LINE messenger, a service used and loved in over 190 countries, and from its hyperscale data center, a sustainable facility run with the utmost security levels. We aim to turn NAVER's technology into a global integrated technology platform that anyone can easily use.

1999 Begins IT operation

2002 Opens Mokdong IDC

2003 Opens Bundang IDC

2005 Opens Yeoksam IDC

2007 Starts free vaccine "NAVER Vaccine"

Opens Mabuk IDC

2008 Opens Trouble Ticket System (TTS),

IDC Facilities Monitoring System (FMS)

2009 NBP founded

2010 Opens Gasan IDC

Opens IP base location information system, NLiveCast and IT service management system (Ntree)

Introduces Candidate Pool System

2012 Opens NCloud Internal

2013 Opens Chuncheon "GAK" Data Center

2014 Introduces NAVER Server G1

Starts NAVER CDN service

Assets due diligence using RFID

2015 Starts International corporations (Asia Pacific, Europe, USA)

Greenpeace Eco-friendly/Sustainable Campaign Evaluation/Assessment

Receives top IDC level

Holds "Hello, Infra", a technology event for affiliate companies

2016 Begins Global Office Care program to supp ort international office work

2017 Starts cloud service (NAVER Cloud Platform)

Opens Peongchon IDC

NAVER "GAK" Data Center receives Energy Champion title from Korea Energy Agency

Enters Japan cloud market

2018 CSAP (Cloud Security Assurance Program) Certified

CSA STAR Gold Certified

ISO27799 Certified

NAVER Cloud Platform enters finance and public industries

Awarded the Minister of Science and ICT award for merit of industrial development

2019 Passes finance security assessments

(Satisfied 109 basic protective measures along with 32 additional protective measures)

2020 Korea Education and Research Information Service (KERIS) adopts NAVER CLOUD infrastructure for online education system

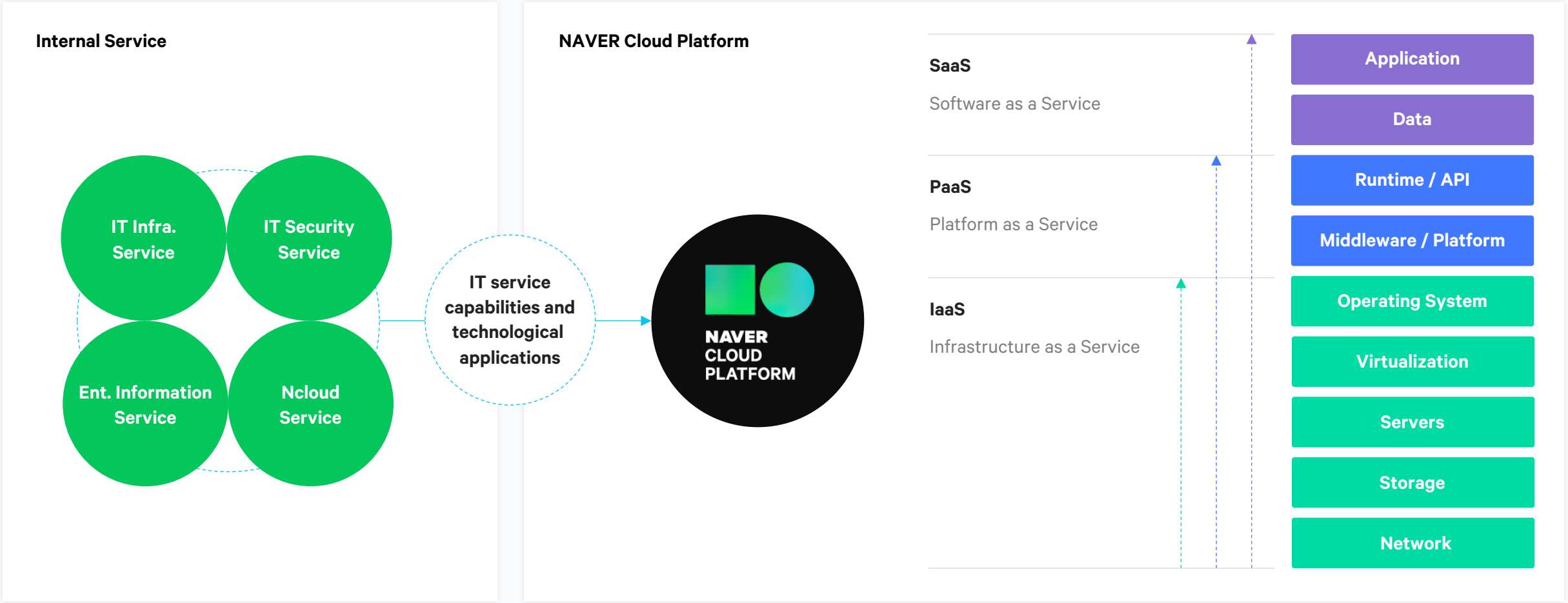
Hyperscale Sejong GAK Data Center set for construction

Launches fully managed private cloud "Neurocloud"

Company name changes to "NAVER CLOUD"

Business

NAVER Cloud is divided into two business areas: NAVER Cloud Platform, a public cloud service, has been growing exponentially each year since its creation in 2017, and Internal Service, which has been built and is operating stably for NAVER and its affiliates since 1999. These two divisions produce a synergy through the integration of each one's capabilities and technologies.



Internal Services

From Korea's #1 NAVER portal to popular BTS's VLIVE live broadcasting,

NAVER Cloud operates services stably for global users and partners to easily use and enjoy.

NAVER

Korea's #1 Internet portal

NAVER continuously grows while adding new value and providing new opportunities to users and partners.

Global Monthly Active Users
2017.12
42 Million

NAVER LABS

LABS, taking on NAVER's future technology

- AI, robotics, automatic driving, mobility, etc.
- Automatic driving guide robot "AROUND G", robot arm "AMBIDEX"
- AI research facility "NAVER LABS EUROPE"

WEBTOON

Global Webtoon content platform

WEBTOON provides new enjoyment through the support of global creators as well as the creation and exchange of topnotch content.

Global Monthly Active Users
2020.08
67 Million

VLIVE

Global celebrities live streaming platform

VLIVE brings global celebrities and fans together, live broadcasts BTS Wembly concert, and builds a Vietnam music chart.

Global Accumulative Downloads
2020.06
93 Million

SNOW

AR camera platform loved around the world

SNOW consistently develops camera-related technologies and unique products and content while creating innovative items.

Global Monthly Active Users
2020.07
242 Million

LINE

Global mobile messenger, portal platform

LINE, a global smart portal, connects people to businesses and services to bring everyday happiness.

Global Monthly Active Users
2020.07
185 Million

NAVER CLOUD PLATFORM
More than doubling in sales, product number and customers every year,
NAVER Cloud’s public cloud service is rapidly evolving.

Service Map

From necessary infrastructure, a hybrid cloud fit to each company's characteristics, AI developed from NAVER's data, company security systems to work collaboration tools, NAVER Cloud Platform continues to grow.

Compute
Discover optimal computing resources from advanced infrastructure and extensive experience

Network
Set up a fast, stable network environment for uninterrupted service, anytime, anywhere

Security
Protect your data and services from external threats with world-class security technologies

Management
Monitor and manage your IT infrastructure and cloud services in real time using stable services

Business Application
Utilize cloud-based company information system and a work collaboration tool for growing companies

Media
Easily and quickly convert high quality media content and build a service platform

Game
Utilize SDK and service linkage support for various essential additional services necessary for game development

Hybrid & Private Cloud
Utilize the various types of hybrid cloud environments to meet the needs of companies

Database
Utilize diverse data platforms including RDBMS, MySQL, Big Data processing and analysis, memory cache, and more

Global
Build safe infrastructure with the global services and data centers in major regions around the world

Analytics
Utilize effective data collection and integrated analysis services of service and system data

Storage
Utilize flexible and secure storage products including Archive Storage, Object Storage, and Backup

AI Service
Choose from the AI services developed from the abundant data of NAVER, LINE, and other affiliates

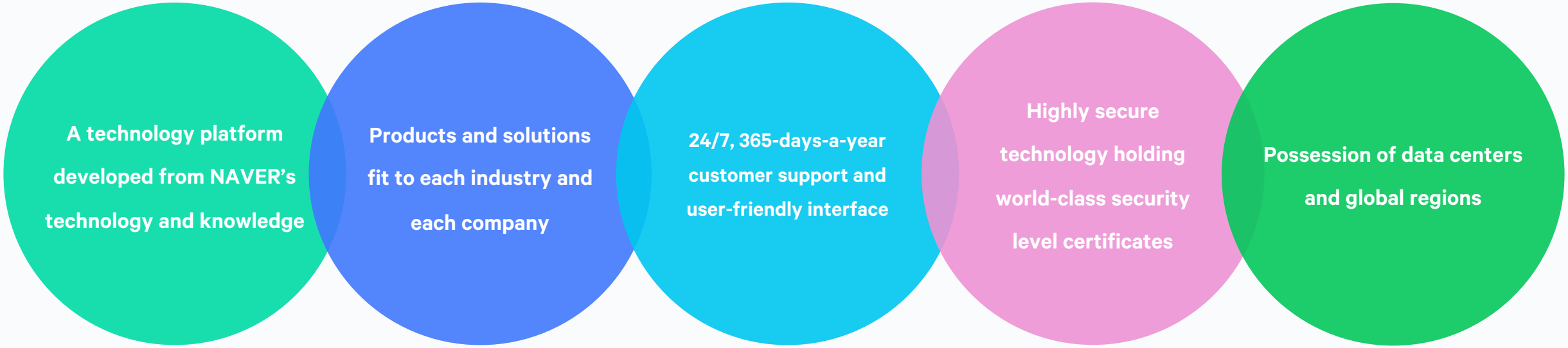
Application Service
Develop diverse solutions from NAVER's services such as Maps and CAPCHA

IoT
Connect millions of devices with cloud and obtain insight safely

Dev Tools
Construct and deploy S/W development environments quickly and securely with DevOps

Blockchain
From private blockchain to blockchain-based services, implement them all on the cloud

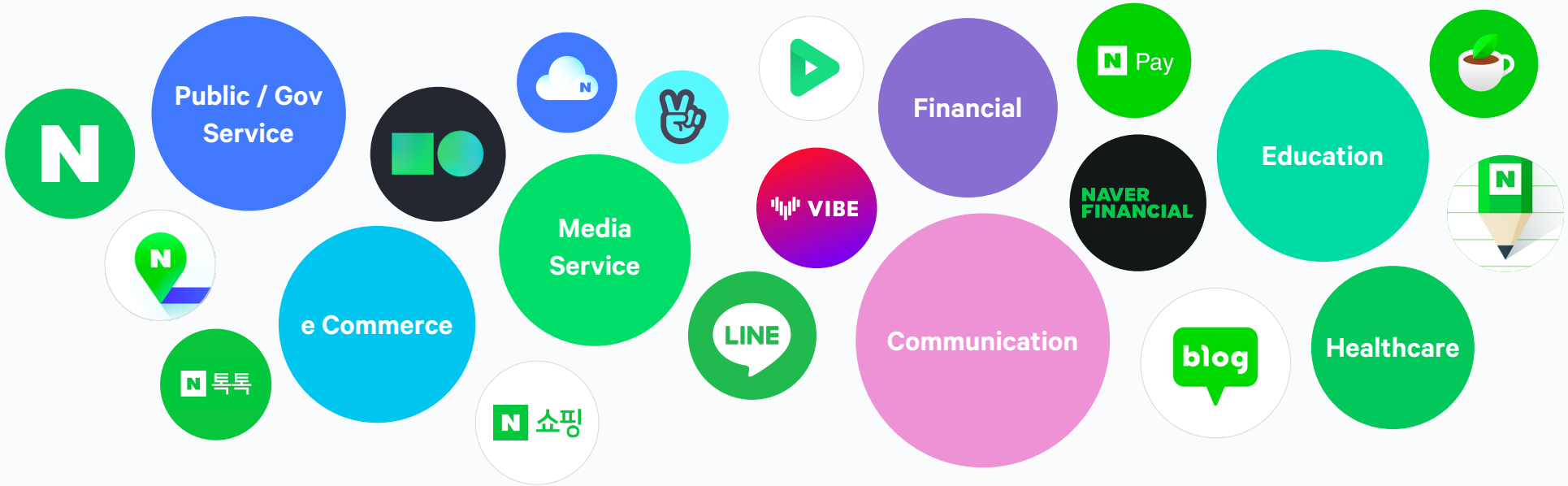
Strength & Value



A technology platform consolidated with NAVER's technology and knowledge

Korea's #1 portal service, NAVER, has diverse content and technology containing abundant data that can be used for various customers' services in a myriad of ways.

We have integrated NAVER's knowledge and technology to be easily utilized in your business.



01

Business Perspective

We provide knowledge and insight of NAVER's diverse service models to be used by businesses.

02

Data Analytics Perspective

We have data collected by NAVER over the years to help analyze and understand users' propensities and trends to then use in one's service.

03

Technological Perspective

We have proven architecture that customers can consult in configuring their digital technology and infrastructure.

Presenting products and solutions fit to each industry and company's growth and success

We rapidly propose cloud-based solutions customized to each industry and company, from startups to large enterprises.

Cloud Service Customized to Industry Type

For the Public Sector

We provide infrastructure that meet security and deliberation requirements as well as AI and Chatbot to build stable public services and convenient civilian services.

For Finance

We provide flexible adoption of reliable services and technologies that comply with the strict regulations required by financial and fintech companies.

For Healthcare

We provide vigorous protection based on high security certifications, secure infrastructure, and novel technology for smart healthcare services.

Neurocloud – Hybrid Cloud

While complying with security regulations, the public cloud can be flexibly linked with legacy environments, offering benefits of both public cloud and on-premises.

Not only does this allow continuous responses to updated data-related regulations, but also to new technologies as well.

Complete Management Service

NAVER Cloud Platform manages everything, from hardware to cloud, lifting up customers' burdens.

On-Premise Linkage

Diverse hybrid cloud structures can be built and linked to the design of the customers' legacy environment.

Data Security Compliance

Because data is processed in-house, you can enlarge your flexible IT environment while complying with regulations.

Diverse Cloud Images

We only provide carefully selected OS and Application images that have been thoroughly verified.

Simple Management Tools

Cloud monitoring and various account systems can be implemented without additional external solutions.

Customizable Service

By providing two types of Neurocloud, customers choose and receive services that fit their situation.

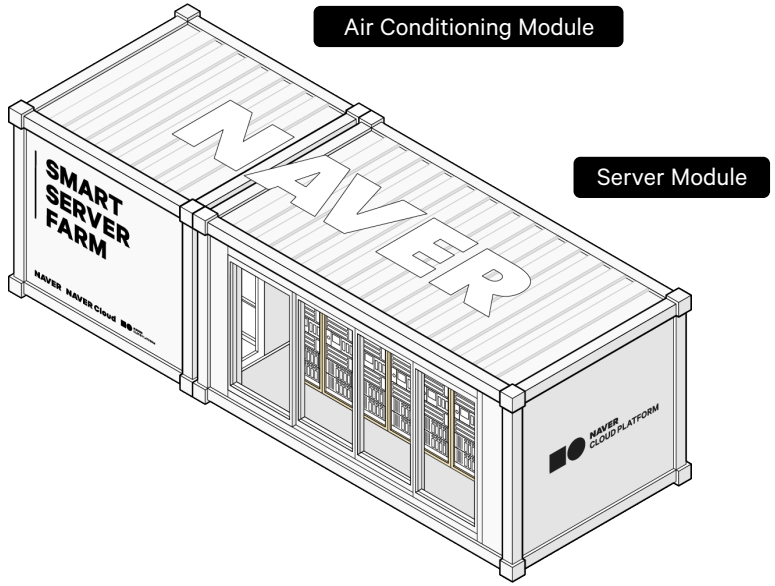
Presenting products and solutions fit to each industry and company's growth and success

We rapidly propose cloud-based solutions customized to each industry and company, from startups to large enterprises.

Neurocloud – Hardware Type

Rack Type

Data Center Type



24/7, 365-days-a-year customer support and user-friendly interface

We have a customer support system that quickly and accurately listens and solves customer problems. And with a user-friendly interface, anyone can easily utilize and manage the service.

NAVER Cloud Platform response lead time an average of 3 hours and 35 minutes

We run a customer service system 24 hours a day, 365 days a year to address customers' issues quickly and flexibly. Contact us via phone or in writing anytime and anywhere a problem arises.

*Average response time to web inquires per business day (in 2019)

NAVER Cloud Platform is fast, friendly and transparent.

- **We are fast.**

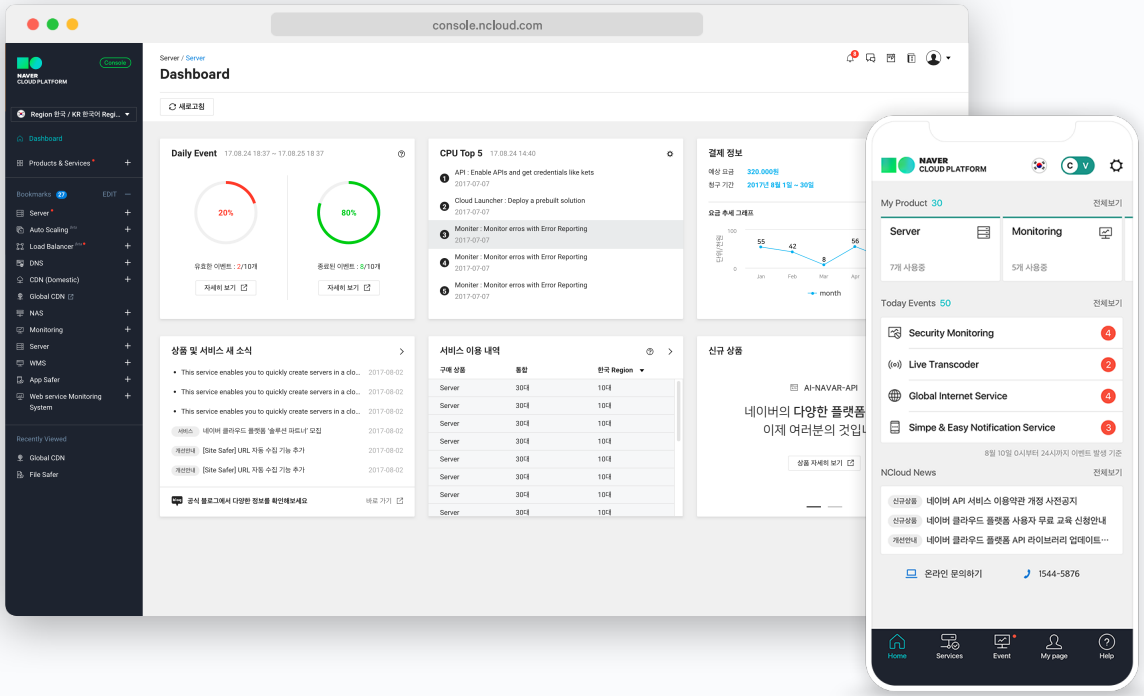
  We work to respond to inquiries on the same day if the issue does not take time for internal verification. We systematically manage a 24-hour customer support process to enable quick response.
- **We are friendly.**

  We are always trying to put ourselves in our customers' shoes. Professional technical staff in each field work day and night to respond accurately and quickly to customer inquires. Should you have any difficulties using the service, we are here to guide you with care.
- **We are transparent.**

  Even after issues are handled, we strive to provide information on the cause and response history to each customer within a few hours. Through various channels such as the portal, e-mail, and text message, we quickly share details related to customers' services.

We provide web and mobile consoles created with the user in mind.

Utilize a straightforward and simple web console and a mobile console that allows service monitoring anytime and anywhere



World-class technology security certifications

We provide a strong and secure cloud that has acquired the world's top security certifications and information protection management system certifications.

Customers can protect data within the country to secure data sovereignty and respond to complex compliances.

Domestic and International Security Certification Acquisition Status

Location	Certification	Badge	Description
Korea	PIMS		Personal information management system certification
	ISMS		Information security management system certification
	Cloud Insurance Assurance Program (CSAP)		Certification for private cloud use by public institutions
Global	ISO/IEC 27001		Global security management standard
	ISO/IEC 27017		Global cloud security and control standard
	ISO/IEC 27018		Global cloud privacy standard
	SOC 2, 3 Report		Global service control level certification
	PCI DSS		Global credit card security standard
	CSA CCM		Global Cloud Security Alliance Cloud Control Matrix

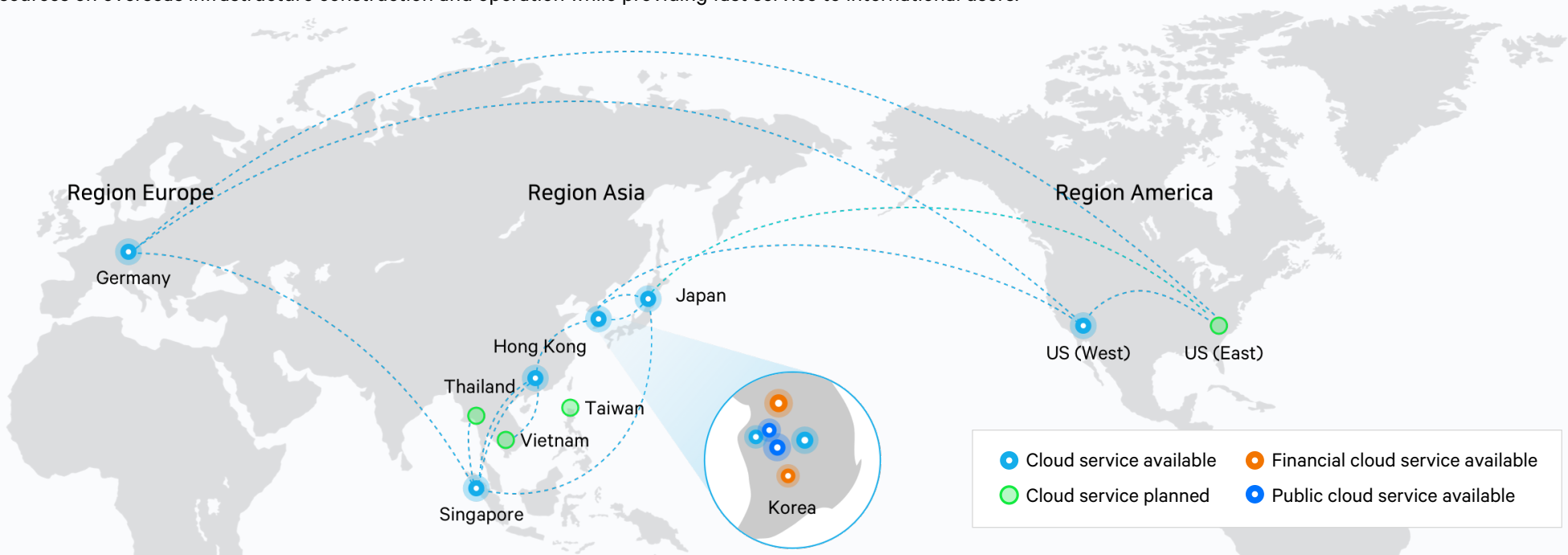
Information Protection Management System Certification Status

Classification	Current Status of Certification		
Information Protection Management System		ISMS	Information security management
		ISO27001	Global Certification of Information Security Management System
Cloud Security		CSAP	Cloud computing service protection (Public cloud certified)
		ISO27017	Global cloud security standard
		ISO27018	Global Cloud Personal Information Security Protection
		CSA STAR	Global cloud control level standard (First in Korea to receive this certification)
Overseas Security Standards		PIMS	Personal information security management system
		SOC2, 3	Global service control level standard
		PCI-DSS	International payment card industry information protection standard

Possession of data centers and global regions

NAVER Cloud is the first and largest portal company in Korea to operate its own data center and services worldwide through regions established internationally.

Customers can save resources on overseas infrastructure construction and operation while providing fast service to international users.



Korea's Cloud Data Center Status

- Operating NAVER Cloud in 2 out of 4 IDCs domestically
- Private Cloud Zone: Operating 2 data centers in a multi-zone format
- Public Cloud Zone: Operating in the form of a main data center and a DR data center
- Financial Cloud Zone: Opened main data center, constructing DR data center

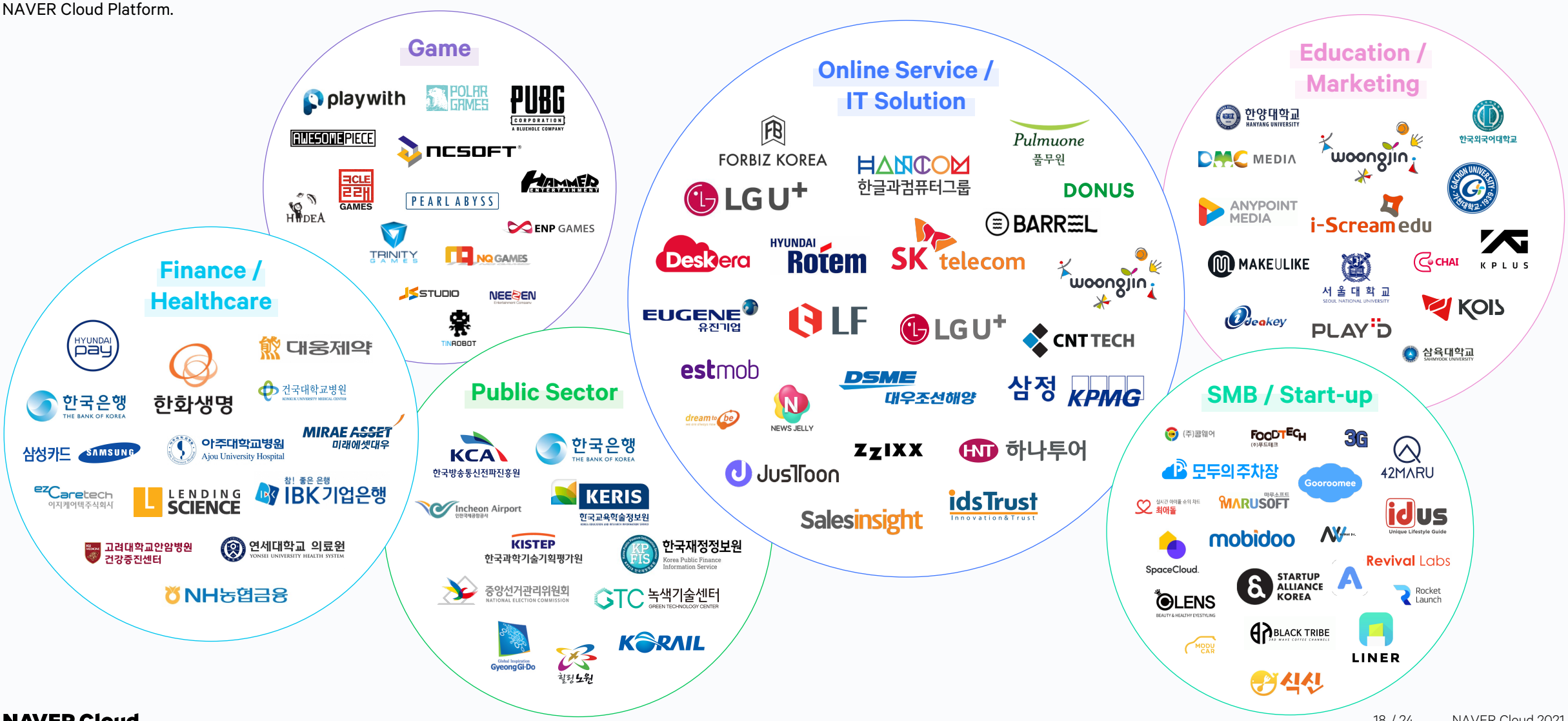
Global Cloud Data Center Status

- Data center operating in 10 regions
- Korea, Japan, Singapore, US East/West, Germany, Hong Kong, Thailand, Taiwan, Vietnam
- Private communication through dedicated line connection between global regions
- Planning to add additional cloud regions

Reference

* Interested in learning about more case studies? Check out www.ncloud.com/intro/cases for additional references.

Customers of various fields and sizes, from large companies to startups and private entrepreneurs and in diverse industries ranging from IT services, gaming, finance, public, education and to media, have selected NAVER Cloud Platform.



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Korea's first life insurance company

Hanwha

Application Method

- **Construction of a core system in the larger insurance system using the hybrid cloud**
- **Utilized in introducing "Claim AI Automatic Examination System"**
  - Establishment of CaaS (Cloud as a Service) in Hanwa Life Insurance data center
  - Establishment and operation of on-premise environment and DB duplexing non-stop service
  - Development as container environment based on MSA* for all basic businesses

*MSA: Micro Service Architecture / the first insurance company in Korea to do so

Results

₩10 mil
Established insurance payment screening fee savings goals*

100%
Stability evaluation satisfaction**

Korea's First
Transfer of core business system to cloud

[Cloud AI Review Introduced]

* Target amount for the next 5 years / ** 109 "basic protection measures", 33 "additional protection measures in the financial sector"

"Making life more valuable through various community platforms"

SAMSUNG CARD

Application Method

- **Application of Chatbot AI in pet community [Ajinyangi]**
  - Addition of natural language search function (using machine learning-based algorithm)
  - Application of exclusive music service for pets linked to CLOVA AI speaker
- **Application of voice recognition in seniors community [Life Doodle]**
  - Application of voice to text changing CLOVA Speech Recognition technology

Results

1-2 day → 1 sec
Health/Nurture/Behavior Question Response Time

[Previously existing 1:1 inquiry ⟶ Chatbot replacement]

4.5/5
Google Play user review*

* Based on "Ajinyangi" App Cumulative Review (2020.08)

One of the top 3 telecommunication companies in Customer Satisfaction

SK telecom

Application Method

- **Application of global region to international roaming service [Baro]**
- **Provision of free voice calls to SKT customers in around 170 countries***
  - Utilization of Global Region infrastructure with major bases around the world
  - Application of Bare Metal Server, Storage, VPN
  - Application of Cloud Connect, Linage of on-premise environment

* When using Wi-Fi or roaming data plan (limited areas)

Results

₩40 mil
Reduction in SKT customer communication cost*

80,00K
Baro international call count*

4.7/5
Google Play user review**

* Based on release of one year cumulative data (Dec. 2019) / ** [T Phone] Based on cumulative app reviews (Aug. 2020)

Reference

* Interested in learning about more case studies? Check out www.ncloud.com/intro/cases for additional references.

Quasi-government organization under the Ministry of Education

KERIS 한국교육학술정보원 KOREA EDUCATION AND RESEARCH INFORMATION SERVICE

Application Method

- **Construction of cloud-based digital education service (video/text)**
- **Smoothly connected 90K people to online schooling (April '20)**
  - [LB-WEB-WAS-DB] Operating a 12 set system expansion
  - Application of Web Monitoring Service (WMS) for quality and secure response
  - Provision of smooth content through CDN service and ISP bandwidth expansion

Results

40 K → 3000 K

Maximum number of possible connections

0 [ZERO]

Server down

Next generation precision medical hospital information system

KU MC KOREA UNIVERSITY MEDICAL CENTER

Application Method

- **Construction of cloud-based [Precision Medical Information System]**
  - Ability to use the same system for each hospital environment modification and usage
  - Enhanced security environment for safe storage of medical data
- **Utilization of high quality medical data for full-fledged precision medicine**
  - Applied customized treatments for patients, new drug development, genome analysis, etc.
  - Established consistent medical big data using standardized medical terms/codes

Results

2 years

Test duration and proven stability

100 hospitals

Sequential introduction scheduled (first half of 2021)*

Korea's First

Senior general hospital to switch to cloud

* 3 hospitals affiliated with Korea University Medical Center (Anam Hospital of Korea, Korea University Guro Hospital and Ansan Hospital of Korea)

Developer of *BATTLEGROUND*

PUBG CORPORATION A BLUEHOLE COMPANY

Application Method

- **Providing [BATTLEGROUND] using Bare Metal Server**
  - Provision of high performance physical service that can be used independently in the form of cloud
  - Suitable for game services that require fast response speed
- **Safe operation of service even while connecting 350K people at once**

Results

Physical Server + Cloud

Utilization of both high performance physical servers advantages and cloud methods at once

5 th place

Korea's game industry revenue (2019)*

* Record for a single game (*BATTLEGROUND*) (PC/console/mobile)

Reference

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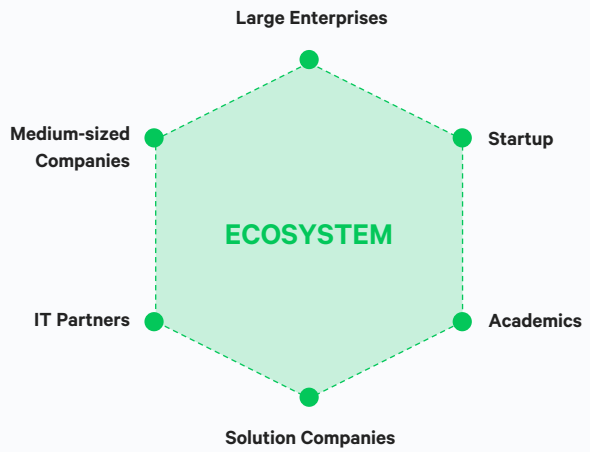
KCA Korea Communications Agency	**Korea Communications Agency** Using public cloud and converted entire system onto cloud	mobidoo	**Mobidoo** A fintech company based on inaudible soundwave transmission technology that utilizes global region services
KORAIL	**KORAIL** Converted internal work system to cloud and connected to personal data center	MIRAE ASSET 미래에셋대우	**Mirae Asset Daewoo** Introduced of HPC for calculating derivatives
NATIONAL ELECTION COMMISSION	**National Election Commission** Converted its homepage infrastructure to cloud	삼정 KPMG	**Samjong KPMG** Using OCR to develop machine recognition program for financial transaction inquires
PEARL ABYSS	**Pearl Abyss** Developer of MMORPG Black Desert Mobile provides its service using CDN	Pulmuone	**Pulmuone** An organic foods distribution company that has built an integrated mall using Kubernetes service
Hanmi	**Hanmi Pharmaceutical** Adopted OCR to automate smart plant document management	BARREL	**Barrel** Held successful online promotion campaign despite increase in large-scale traffic thanks to the introduction of cloud

Ecosystem

We aim to build a virtuous cloud-based ecosystem to help the growth of all members of the IT ecosystem. To achieve this, we operate various training and support programs.



Success is made together.

Through collaborating with competitively advantageous solution companies and IT partners, NAVER Cloud provides startups and small businesses with support programs. We will constantly grow and strive to achieve our goals together.

Greenhouse Benefit

NAVER CLOUD PLATFORM

NAVER Cloud Platform Startup Support Program

NAVER Cloud Platform's Greenhouse Program actively provides startups with cloud infrastructure training and technology consulting necessary for growth. Currently, over 2,000 startups are in the program. We strive for mutual growth through active joint marketing with startups.

[Information on affiliates and detailed benefits]
www.ncloud.com/support/greenHouse

Green Rookie

NAVER CLOUD PLATFORM

NAVER Cloud Platform Academic Support Program

We provide cloud infrastructure and training for future IT professionals who wish to expand their dreams on cloud. We offer additional cloud benefits to students and faculty members who partner with NAVER Cloud Platform Green Rookie Program, which will be used more in schools and in society in the future.

[Information on affiliated educational institutions and detailed benefits]
www.ncloud.com/support/greenRookie

Break Through with

NAVER Cloud

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Thank You

NAVER Cloud